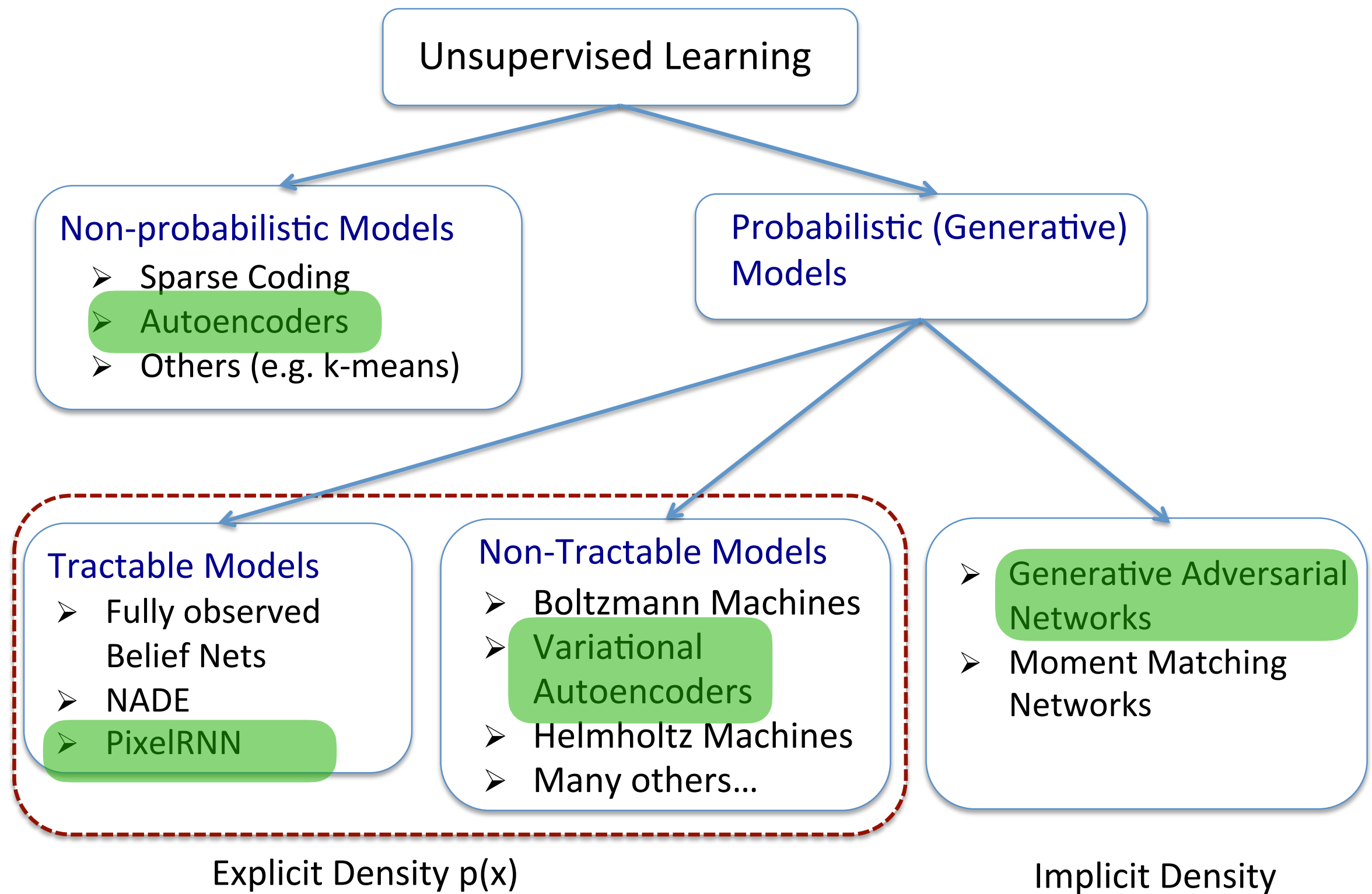


Deep Learning with Neural Networks

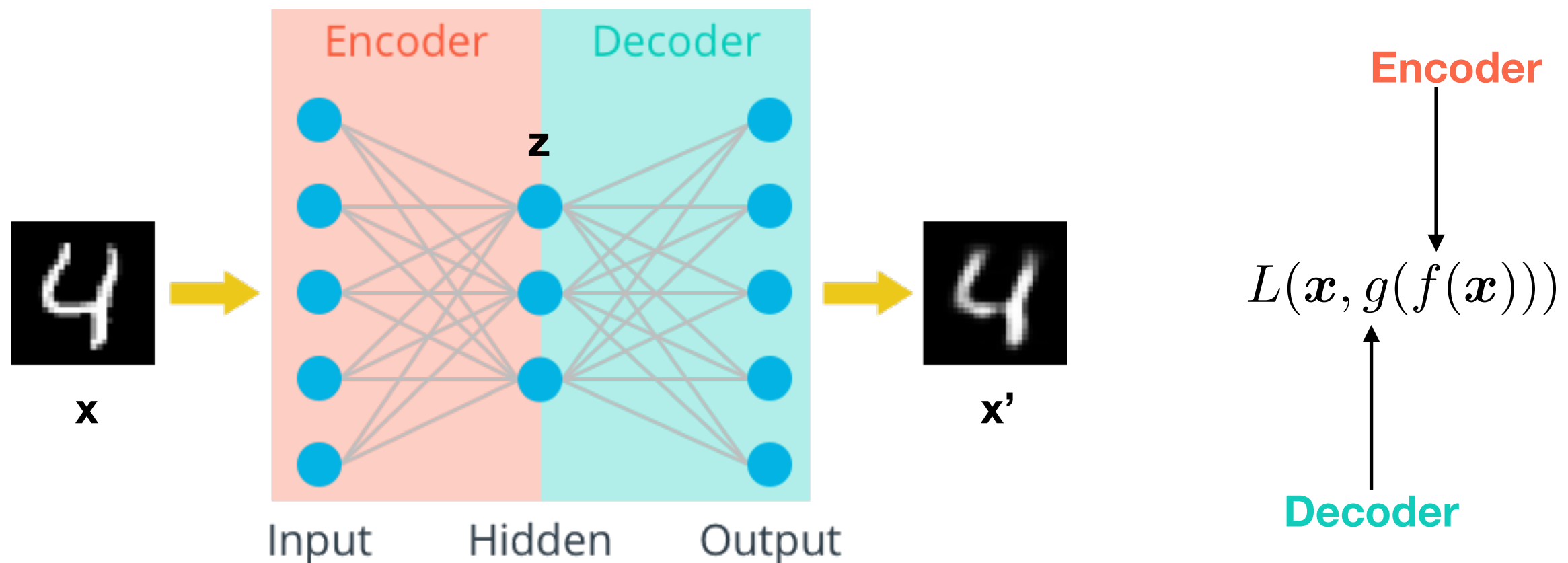
Unsupervised Deep Learning (I): autoencoders

Original: Pablo Martínez Olmos, pamartin@ing.uc3m.es
Adapted: Alejandro Guerrero López, alguerre@pa.uc3m.es



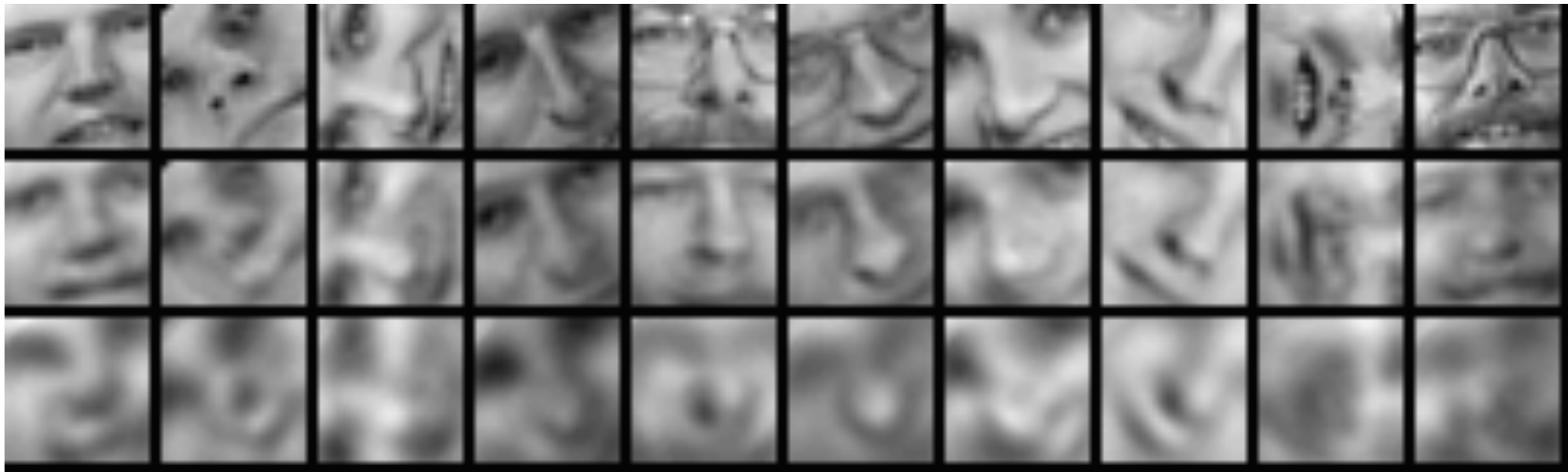
Deep Autoencoder

- An autoencoder is a neural network that is trained to attempt to copy its input to its output
- Internally, it has a low-dimensional hidden representation
- Two parts: encoder $f(\mathbf{x})$ and decoder $g(\mathbf{z})$:
 - where $\mathbf{x}$ is the input data, e.g., an image of a four
 - where $\mathbf{z} = f(\mathbf{x})$, i.e., the output of the encoder
- Autoencoders have been traditionally used for dimensionality reduction or feature learning



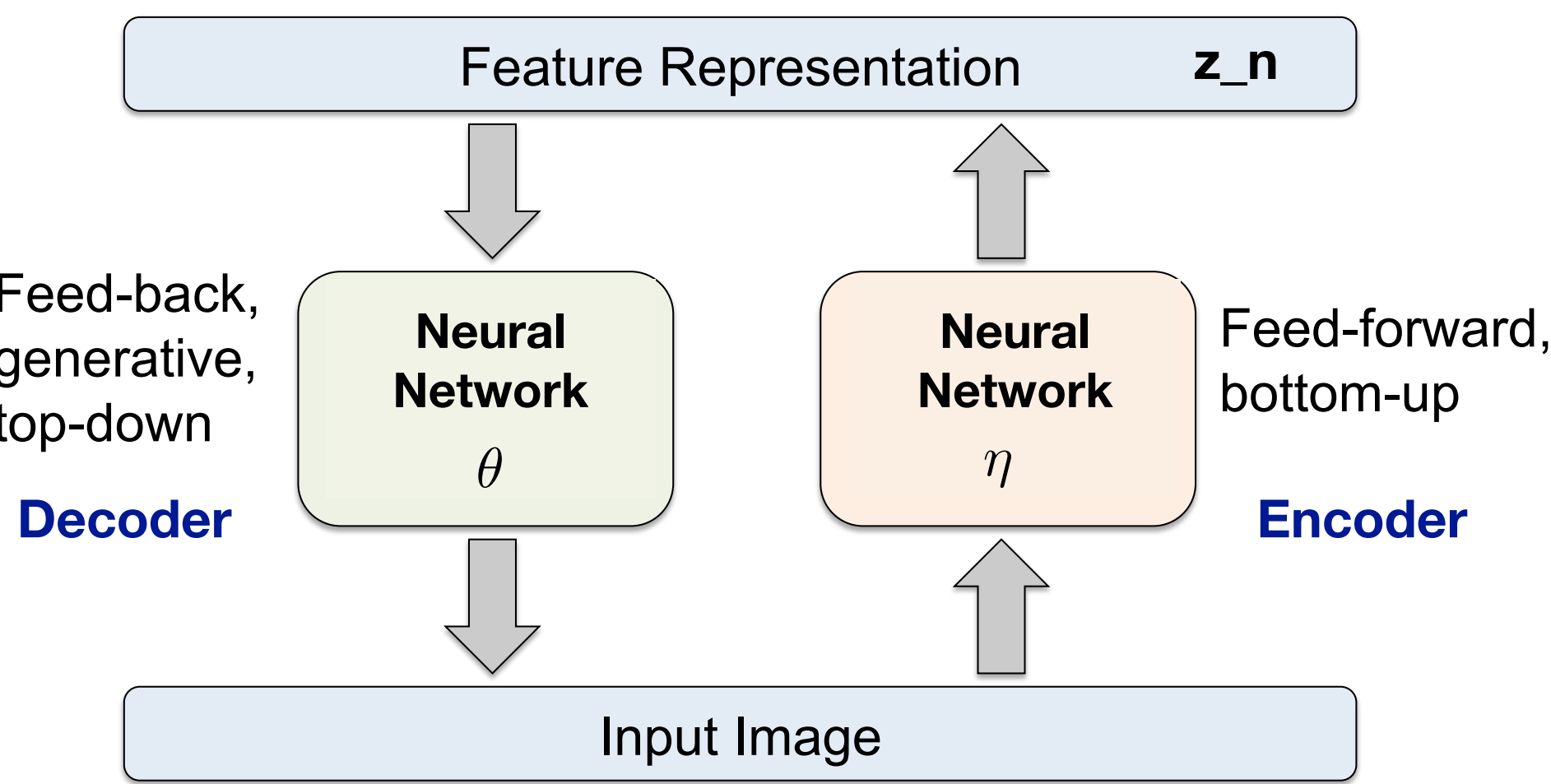
Deep Autoencoder

- 25x25 – 2000 – 1000 – 500 – 30 autoencoder to extract 30-D real-valued codes for Olivetti face patches.



- **Top:** Random samples from the test dataset.
- **Middle:** Reconstructions by the 30-dimensional deep autoencoder.
- **Bottom:** Reconstructions by the 30-dimensional PCA.

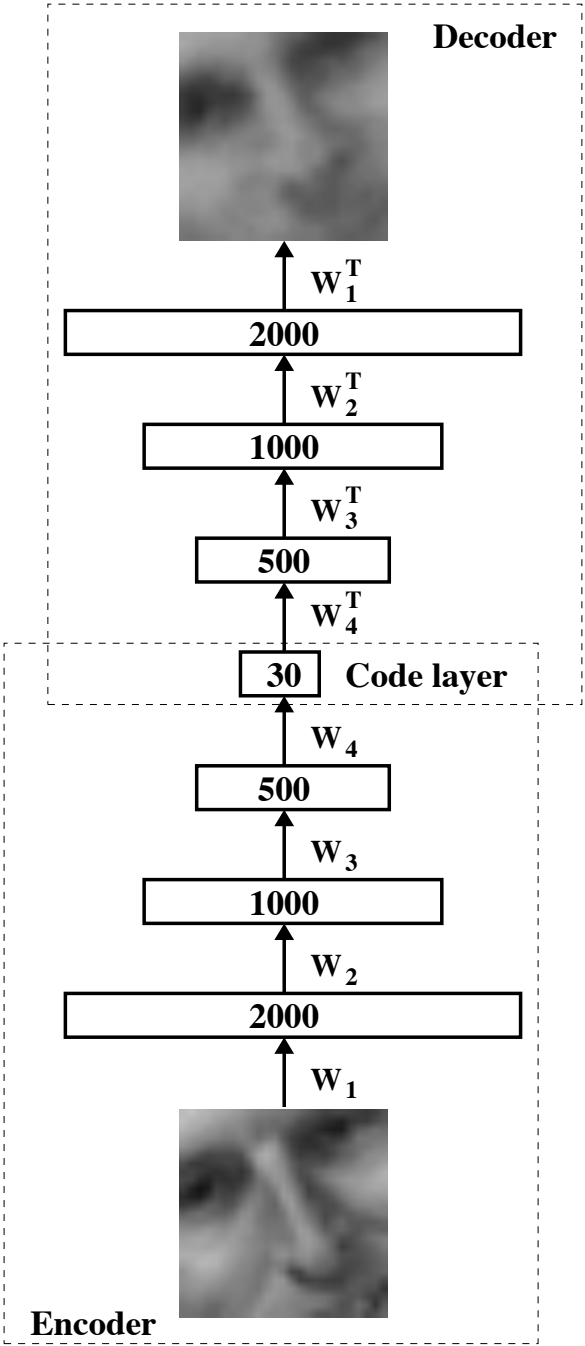
Deep Autoencoders



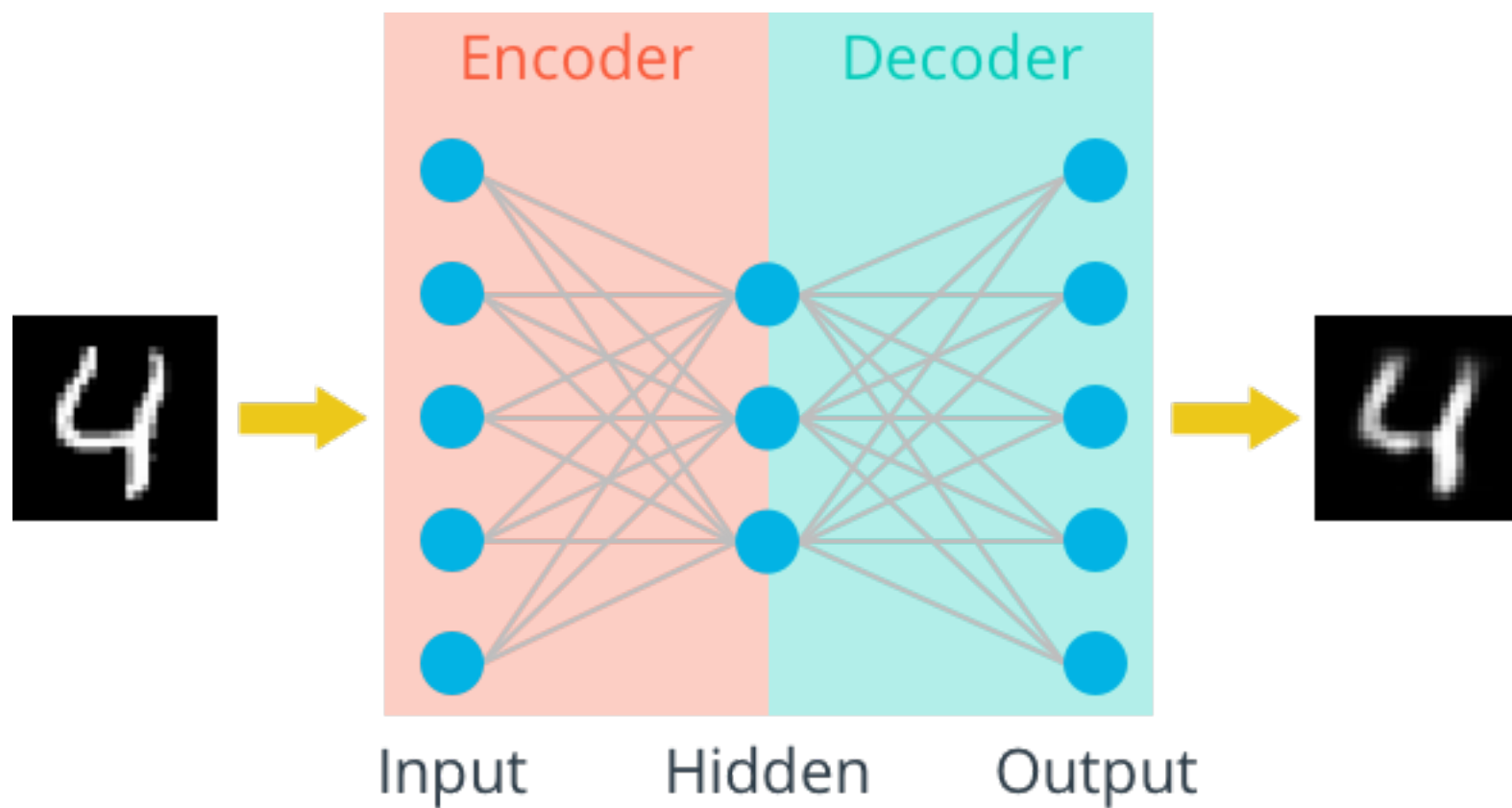
Encoder

$$\mathcal{L}(\eta, \theta) = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_n (\mathbf{x}_n - D_{\theta} (E_{\eta}(\mathbf{x}_n)))$$

Decoder



Sparse Deep Autoencoders



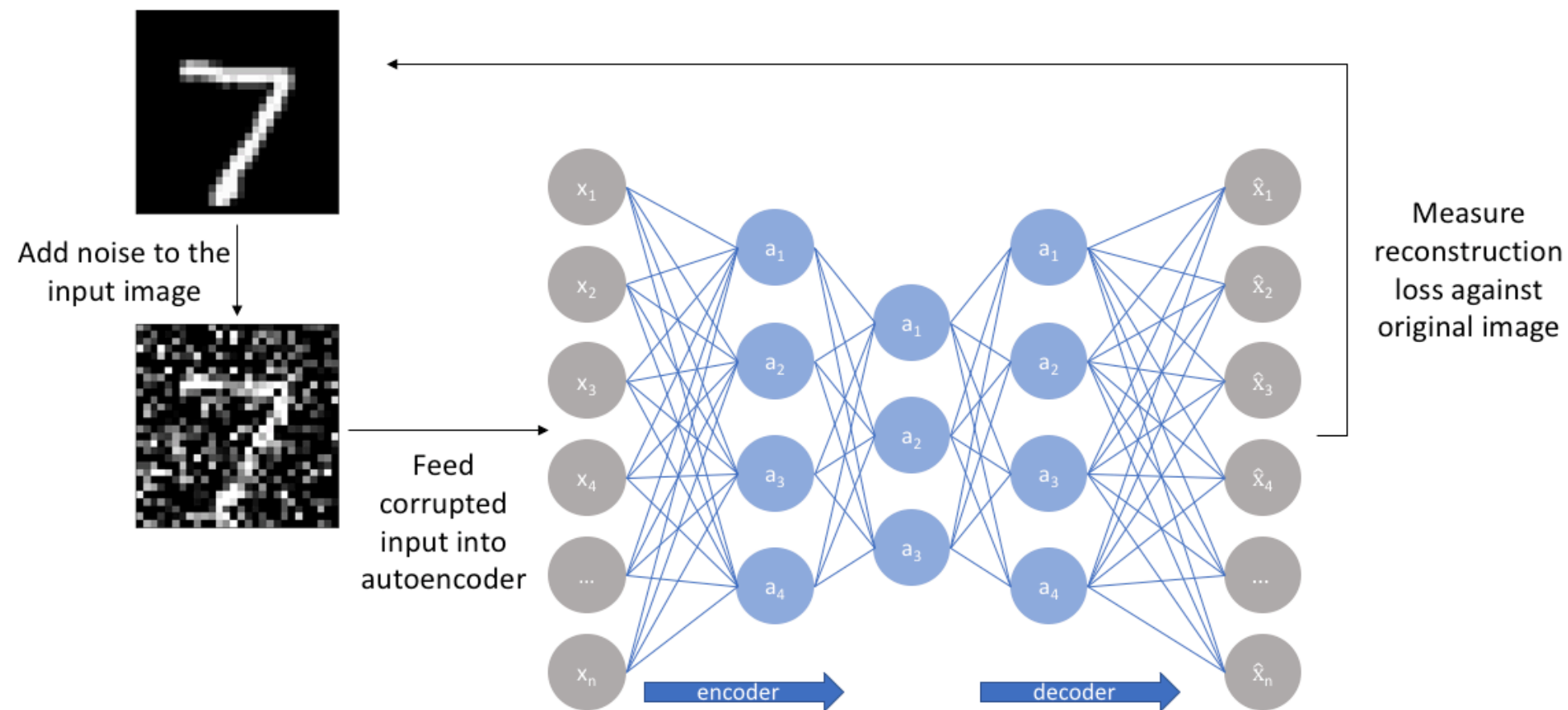
$$L(\mathbf{x}, g(f(\mathbf{x}))) + \Omega(\mathbf{h})$$

The diagram shows the loss function for the Sparse Deep Autoencoder. The **Encoder** maps the input $\mathbf{x}$ to the hidden representation $\mathbf{h}$ via the function f . The **Decoder** maps the hidden representation $\mathbf{h}$ back to the output $g(\mathbf{h})$ via the function g . The loss function is the sum of the reconstruction loss $L(\mathbf{x}, g(f(\mathbf{x})))$ and the sparsity penalizer $\Omega(\mathbf{h})$. The term $\Omega(\mathbf{h})$ is highlighted with a green circle and a green arrow pointing to it from the text 'Sparsity penalizer' below.

$$\Omega(\mathbf{h}) = \lambda \sum_i |h_i|$$

- Sparse autoencoders are typically used to learn features for another task

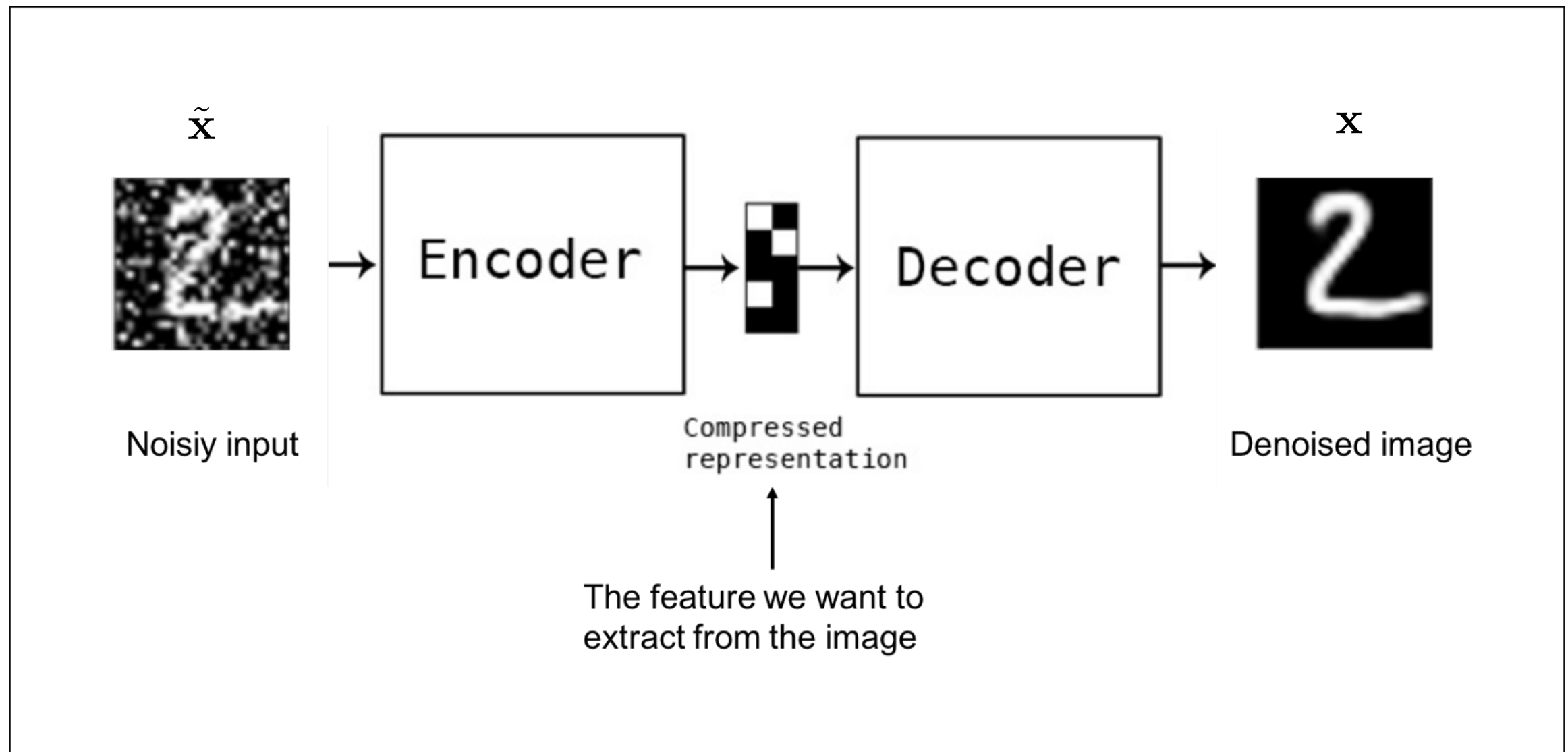
Denoising Deep Autoencoders



Source: [this excellent blog](#)

- Reconstruct a **corrupted version** of an image
- More robust solutions. It is some sort of regularization
- They are widely used for image denoising and missing data completion

Denoising Deep Autoencoders



$$\mathcal{L}(\eta, \theta) = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_n(\mathbf{x}_n - D_{\theta}(E_{\eta}(\tilde{\mathbf{x}}_n)))$$